

MMATHS 2025 Individual Round Solutions

Yale Math Competitions

November 2025

1. Lily writes 10 consecutive two-digit numbers on a sheet of paper. Given that the digit 1 appears seven times, how many times does the digit 2 appear?

Proposed by: Jason Mao

Answer: $\boxed{5}$

Solution: Since there are 10 consecutive numbers, each digit 0 through 9 inclusive will always appear exactly once in the ones place. Hence, the fact that there are 7 ones means there are 6 ones in the tens places. For two digit numbers, this can only occur with the numbers 10 to 19, and hence the other numbers in the set must be in 20 to 29, meaning there will be 6 numbers with a one in the tens place, 4 numbers with a two in the tens place, and thus $4 + 1 = \boxed{5}$ twos.

2. How many ordered pairs of real numbers (x, y) satisfy the equation

$$(x + y - 1)^2 + (x^2 + y^2 - 1)^2 = 0?$$

Proposed by: Patrick Lu

Answer: $\boxed{2}$

Solution: By the trivial inequality, $(x + y - 1)^2 \geq 0$ and $(x^2 + y^2 - 1)^2 \geq 0$, so the only way the sum can equal zero is if both are zero, meaning $x + y = 1$ and $x^2 + y^2 = 1$. This means $0 = 1^2 - 1 = (x + y)^2 - (x^2 + y^2) = 2xy$, so $xy = 0$, and hence either x or y is equal to 0, and since they sum to 1, the other equals 1. The possible solutions are therefore $(0, 1)$ and $(1, 0)$, which gives a final answer of $\boxed{2}$.

3. Let $ABCD$ be a parallelogram with $\overline{AB} \parallel \overline{CD}$ and $\overline{AD} \parallel \overline{BC}$. Extend $\overline{AD}$ past D to meet the angle bisector of $\angle ABC$ at a point E . If $AB = 55$ and $BC = 45$, what is the length of $\overline{DE}$?

Proposed by: Patrick Lu

Answer: $\boxed{10}$

Solution: Since $\overline{AD}$ is parallel to $\overline{BC}$, we have that $\angle EBC = \angle AEB$. Moreover, since $\overline{EB}$

is a bisector, we have $\angle ABE = \angle EBC$. Thus $\angle ABE = \angle AEB$, and so ABE is isosceles with $AB = AE$. Thus $55 = AD + DE \implies DE = 55 - 45 = \boxed{10}$

4. Cat and Claire are discussing their favorite two-digit positive integers.

Cat: Do you think 49 is a divisor of the product of our numbers? I can't tell.

Claire: I can't tell either.

Cat: Oh, now I'm sure the product is divisible by 49.

Claire: Interesting! That means the sum of our numbers can't possibly be divisible by 11.

Cat: The sum of our numbers is definitely divisible by 13, though.

Let Cat's and Claire's numbers be A and B , respectively. What is $100A + B$?

Proposed by: Jason Mao

Answer: $\boxed{6328}$

Solution: Let's work through the logic line-by-line.

- **Line #1.** This tells us Cat's favorite number isn't divisible by 49, because otherwise, she would be able to tell.
- **Line #2.** Claire knows that Cat's favorite number is not divisible by 49. Therefore, if Claire's number were not divisible by 7, then she would be certain that the product is not divisible by 49. But Claire says she doesn't know! Thus, Claire's number must be divisible by 7, but not by 49.
- **Line #3.** Cat knows that Claire's favorite number is divisible by 7 but not by 49. In order to claim that the product is divisible by 49, Cat must also have a number divisible by 7 but not by 49. Thus, Cat's and Claire's numbers are $7m$ and $7n$ for positive integers $m \neq 7$ and $n \neq 7$ between 2 and 13, inclusive.
- **Line #4.** The sum of Cat's and Claire's numbers is $7(m+n)$. Claire claims that $7(m+n)$ cannot be divisible by 11. In other words, Claire—knowing the value of n —deduces that $m+n$ cannot equal either 11 or 22. Observe that when $n = 4$ (and only when $n = 4$), this deduction is possible! This is because when $n = 4$, the conditions $m+n = 11$ and $m+n = 22$ imply $m = 7$ or $m = 18$, both of which are impossible. Thus, $n = 4$.
- **Line #5.** Cat now knows that Claire's number is 28. The only value of m for which $7m + 28 = 7(m+4)$ is divisible by 13 is $m = 13 - 4 = 9$. Thus, $m = 9$.

In conclusion, Cat's number is $A = 63$, and Claire's number is $B = 28$, so the answer is $\boxed{6328}$.

5. There is a magical bee in Catherine's room! Every minute, magical bees decide between two choices, each with probability $\frac{1}{2}$: either sting Catherine and disappear, or split into three new magical bees, who each then act independently. Thus, after one minute, Catherine has been stung with probability $\frac{1}{2}$ or there are now three bees in the room. The expected number of times that Catherine has been stung after four minutes can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Proposed by: Howard Dai

Answer: $\boxed{81}$

Solution: Let $E(n)$ be the expected number of stings after n minutes. First, let $B(n)$ be the expected number of bees after n minutes, with $B(0) = 1$ and clearly $B(n) = \frac{3}{2}B(n-1)$. Since stings are cumulative and at each minute the expected number of stings added is half the expected number of bees in the previous minute, we can get

$$E(n) = \frac{1}{2}(B(n-1) + B(n-2) + \cdots + B(0)) = \frac{B(0)}{2} + \frac{3}{2} \frac{1}{2}(B(n-2) + \cdots + B(0)) = \frac{1}{2} + \frac{3}{2}E(n-1)$$

Clearly $E(0) = 0$. We can now repeatedly use this recursion (and the given $E(1) = 0.5$).

$$\begin{aligned} E(4) &= \frac{1}{2} + \frac{3}{2}E(3) \\ \frac{1}{2} + \frac{3}{2}E(3) &= \frac{1}{2} + \frac{3}{4} + \frac{9}{4}E(2) \\ \frac{1}{2} + \frac{3}{4} + \frac{9}{4}E(2) &= \frac{1}{2} + \frac{3}{4} + \frac{9}{8} + \frac{27}{8}E(1) \\ \frac{1}{2} + \frac{3}{4} + \frac{9}{8} + \frac{27}{8}E(1) &= \frac{1}{2} + \frac{3}{4} + \frac{9}{8} + \frac{27}{16} \\ \frac{1}{2} + \frac{3}{4} + \frac{9}{8} + \frac{27}{16} &= \frac{65}{16} \end{aligned}$$

so our final answer is $65 + 16 = \boxed{81}$

6. Let a , b , and c be the distinct roots of $2x^3 - x^2 + x - 1$. Find the value of $\sum_{n=1}^{\infty} a^n + b^n + c^n$.

Proposed by: Alex Wa

Answer: $\boxed{2}$

Solution: We begin by adding $a^0 + b^0 + c^0 - 3 = 0$ to the summation so the summation becomes

$$-3 + \sum_{n=0}^{\infty} a^n + b^n + c^n = -3 + \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$$

By combining the fractions, we find this equals

$$-3 + \frac{3 - 2(a+b+c) + ab + bc + ca}{(1-a)(1-b)(1-c)}$$

By Vieta's formulas, we compute that the numerator is equal to $3 - 2(\frac{1}{2}) + (-\frac{1}{2})$. Since the roots of $2x^3 - x^2 + x - 1$ are a, b, c , it follows that $2(x-a)(x-b)(x-c) = 2x^3 - x^2 + x - 1$. So plugging in $x = 1$ gives $2(1-a)(1-b)(1-c) = 1$, so the denominator is $\frac{1}{2}$. Thus our final answer is:

$$-3 + \frac{3 - 1 + \frac{1}{2}}{\frac{1}{2}} = -3 + 5 = \boxed{2}$$

7. How many integers x with $3 \leq x \leq 1000$ satisfy the property that for every power of three 3^k between 3 and x , inclusive, $\frac{x}{3^k} - \left\lfloor \frac{x}{3^k} \right\rfloor$ is at least $\frac{1}{3}$, where $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z ?

Proposed by: Shane Lee

Answer: 124

Solution: Consider x expressed in base 3. The fractional part of $x/3^k$ being at least $\frac{1}{3}$ means that after shifting the decimal point to the left by k places, the first digit after it is 1 or 2. Applying this to all such k means that a necessary and sufficient criterion is that the base-3 representation of x consists entirely of 1's and 2's. Since $x \geq 3$, x must have at least 2 digits. Since $1000_{10} = 1101001_3$, but the smallest 7-digit base-3 number with only 1's and 2's is 1111111_3 , all valid integers x must have exactly 2, 3, 4, 5, or 6 digits in base 3. Since each digit has 2 choices (1 or 2), the total number of such integers is $2^2 + 2^3 + 2^4 + 2^5 + 2^6 = 4 + 8 + 16 + 32 + 64 = \boxed{124}$.

8. Let ABC be an equilateral triangle with side length 2025. The set of all points P that satisfy the equation $PA = PB + PC$ traces out a curve of length $m\sqrt{n}\pi$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Proposed by: Patrick Lu

Answer: 453

Solution: Suppose that A , B , and C are arranged clockwise. Consider any such P . Rotate P by 60° counterclockwise about A to another point P' . Note that

$$PP' = AP = BP + CP = BP + BP',$$

and thus B lies on segment PP' . Thus

$$\angle BPA = \angle P'PA = 60^\circ,$$

and P lies on the circumcircle of $\triangle ABC$. It's clear that P must lie on minor arc $\widehat{BC}$, and all such points satisfy the given equation. The length of $\widehat{BC}$ is $\frac{1}{3} \cdot 2 \cdot \frac{2025}{\sqrt{3}}\pi = 450\sqrt{3}\pi$, giving an answer of 453.

9. Let $\{a_n\}$ be a sequence of nonnegative real numbers such that a_0 is a proper divisor of 2025^{2025} , and for all positive integers n , $a_n^{a_{n+1}} = |2025^{2025} - a_0 n|$. What is the value of $a_0^{a_1}$?

Proposed by: Aron Fekete

Answer: 1

Solution: Consider $k = \frac{2025^{2025}}{a_0}$. Since a_0 is a proper divisor of 2025^{2025} , we have that k is a positive integer and $k > 1$, meaning that $k - 1$ is also a positive integer. First, consider $n = k$. We get that $a_k^{a_{k+1}} = |2025^{2025} - a_0 k| = |a_0 k - a_0 k| = 0$. Since both a_k and a_{k+1} are

nonnegative reals, we must have $a_k = 0$ for this to be possible. Now consider $n = k - 1$. We get $1 = a_{k-1}^0 = a_{k-1}^{a_k} = |a_0 k - a_0(k-1)| = |a_0|$ and since a_0 is nonnegative we must have $a_0 = 1$, so our final answer is $a_0^{a_1} = 1^{a_1} = \boxed{1}$

10. Benjamin has 1000 t-shirts, labeled $1, 2, \dots, 1000$. These labels also correspond to his preferences for shirts: he prefers shirt i to shirt j if and only if $i < j$. On day 0, Benjamin randomly arranges the shirts into two stacks of 500 shirts each. Let a_k denote the k th shirt in the first stack (from the top), and b_k denote the k th shirt in the second stack in this initial arrangement. Each subsequent day, Benjamin looks at the top shirt of each stack and wears the one he likes more, removing it from the stack. Benjamin continues this process until both stacks are empty. Given that Benjamin prefers a_{20} to b_{25} , the probability that b_{25} is worn before a_{20} can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Proposed by: Howard Dai

Answer: $\boxed{64}$

Solution: We only need to consider the set

$$S = \{a_1, \dots, a_{20}, b_1, \dots, b_{25}\}$$

since any shirts below a_{20} or b_{25} cannot affect which of the two is worn first. Now consider the shirt $M \in S$ with the maximum label (the shirt Benjamin likes the least). Since $b_{25} > a_{20}$, $M \neq a_{20}$. If $M \in \{a_1, \dots, a_{19}\}$, then all of $b_1, \dots, b_{25}$ will be selected before a_{20} . Similarly, if $M \in \{b_1, \dots, b_{25}\}$, then all of $a_1, \dots, a_{20}$ will be selected before b_{25} . So the probability b_{25} is selected before a_{20} is, by Bayes' Theorem:

$$P(M \in \{a_1, \dots, a_{19}\} \mid a_{20} < b_{25}) = \frac{P(a_{20} < b_{25} \mid M \in \{a_1, \dots, a_{19}\}) \cdot P(M \in \{a_1, \dots, a_{19}\})}{P(a_{20} < b_{25})}.$$

Since the shirts are randomly distributed, the denominator is $\frac{1}{2}$, and the right term of the numerator is $\frac{19}{45}$. For the left term of the numerator, the position of the maximum does not affect the ordering of a_{20} and b_{25} , so the left term is just $\frac{1}{2}$. So our desired probability is $\frac{19}{45}$ which gives an answer $\boxed{64}$.

11. Let $\sigma(n)$ denote the sum of the positive divisors of n , and let $\varphi(n)$ denote the number of positive integers less than or equal to n that are relatively prime to n . There exists a unique polynomial P such that

$$P(n) \geq 2\sigma(n) + n\varphi(n)$$

for all positive integers n , with equality holding for infinitely many such n . Find $P(10)$.

Proposed by: Aron Fekete

Answer: $\boxed{112}$

Solution: Notice that by symmetry, when $n > 1$, the sum of all integers (define this sum

$f(n)$ less than or equal to n that are relatively prime with n will be equal to $\frac{n\varphi(n)}{2}$, so our expression is equal to $2(\sigma(n) + f(n))$ and the only integer that can be both a divisor of n and relatively prime with n is 1, so

$$2(\sigma(n) + f(n)) = 2(2 + \sum_{1 < i \leq n, \gcd(i, n) \in \{1, i\}} i) \leq 2(2 + \sum_{i=2}^n i) = n^2 + n + 2$$

We achieve equality when all integers less than or equal to n are either divisors of n or relatively prime to n . This occurs when n is prime, which happens infinitely many times, so $P(n) = n^2 + n + 2$ and our answer is $P(10) = \boxed{112}$.

Remark: There is a rather clean way to show that equality only holds when n is prime or equal to 4. Simply let p be the smallest prime divisor of composite $n > 4$, and since n is composite, $n \geq p^2 \geq 2p$ (equality can only hold when $p = 2$ and $n = 4$) so $\frac{n}{2} < n - p < n$ which means $n - p$ cannot be a divisor of n , but it also shares the factor p so it is not coprime either and hence equality cannot hold.

12. Let ABC be a triangle with $AB = 13$, $BC = 14$, and $AC = 15$. Let ω be the incircle of $\triangle ABC$. There exists a circle Γ passing through B and C that is internally tangent to ω at a point P . The line tangent to ω and Γ at P meets $\overline{AB}$ and $\overline{AC}$ at points K and L , respectively. The length of $\overline{KL}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Proposed by: Yibo Zhang

Answer: $\boxed{314}$

Solution: Denote the intersections of ω and AB, BC, AC as R, D, S respectively, let KM meet line BC at Q . Since QP is tangent to ω and Γ , let $QB = x$, $QP = QD = x + 6$, by Power of a Point with respect to Γ , we have $x(x + 14) = (x + 6)^2$, $x = 18$. Now we compute: since the inradius of $\triangle ABC$ is 4, $KP = 4 \tan \angle KIP$, where $\angle KIP = \frac{1}{2}(\angle PID - \angle RID) = \angle IBD - \angle IQD$; similar reasoning yields $\angle MIP = \frac{1}{2}(360^\circ - \angle PID - \angle SID) = \frac{1}{2}(\angle ACD + \angle PID) = \angle ICD + \angle IQD$. We have $\tan \angle IQD = \frac{1}{6}$, $\tan \angle IBD = \frac{2}{3}$, $\tan \angle ICD = \frac{1}{2}$; tangent addition and subtraction yields $\tan \angle KIP = \frac{\frac{2}{3} - \frac{1}{6}}{1 + \frac{1}{6} \cdot \frac{2}{3}} = \frac{9}{20}$, $KP = \frac{9}{5} \tan \angle MIP = \frac{\frac{1}{6} + \frac{1}{2}}{1 - \frac{1}{6} \cdot \frac{1}{2}} = \frac{8}{11}$, $MP = \frac{32}{11}$. Sum up $\frac{9}{5} + \frac{32}{11} = \frac{259}{55}$ to get $\boxed{314}$.